Venkata Lakshmi Parimala Pasupuleti

Data Analyst · Python · R · SQL · Tableau · Power BI · Statistical Modeling · Data Visualization · Research Analytics

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SUMMARY

Data Analyst with M.S. Data Analytics Engineering from George Mason University (GPA 3.87, May 2026) and direct experience supporting faculty research and academic stakeholders at GMU. Built multi-source statistical models ($R^2 = 0.977$ across 190+ countries), designed faculty leadership dashboards in Tableau and ggplot2, and applied anomaly detection to institutional research pipelines. 8 years of data pipeline, financial analytics, and compliance reporting experience at Entrata (100K+ users). Proficient in Python, R, SQL, Tableau, Power BI, and statistical analysis.

EXPERIENCE

George Mason University - Research Associate

Aug 2024 - Aug 2025

Research Associate · College of Engineering and Computing · Virginia

- Built multi-source statistical prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - data collection, cleaning, transformation, and validation pipeline; achieved **$R^2 = 0.977$** regression accuracy.
- Automated data integrity checks across 3 research pipelines by applying Isolation Forest anomaly detection - surfaced anomalies manual review missed, improving downstream model reliability for faculty stakeholders.
- Designed faculty leadership dashboards in **Tableau, ggplot2, and Quarto** - improved research comprehension and reporting quality by **40%** across two cohorts; delivered findings directly to faculty and academic leadership.
- Conducted hypothesis testing, exploratory data analysis, and quantitative analysis driving research strategy recommendations; documented methodology for reproducibility and institutional knowledge transfer.

George Mason University - Graduate Teaching Assistant

Aug 2025 - May 2026

Graduate Teaching Assistant · College of Engineering and Computing · Part-time · On-site

- Supported 600+ students across MIS 303 sections - Python programming, database design, SQL query writing, MS Access, and Excel/pivot table analysis; coordinated content delivery with lead instructor across all sections.
- Graded assignments with rubric-aligned feedback; held weekly office hours for targeted one-on-one analytical and technical support.

Entrata - Senior Software Engineer (Data & Analytics Focus)

Sep 2018 - Aug 2024

Senior Software Engineer · PropTech / Real Estate SaaS · Lehi, UT (Remote)

- Managed end-to-end financial analytics and reporting for affordable housing compliance: HAP allocation, AMI-based pricing tiers, tax credit calculations, HUD payments, rental subsidies, and ledger reconciliation - quantitative risk analysis across 100K+ unit portfolios; zero compliance violations.
- Processed **1.6M HUD and Tax Credit income limit records** across 5 years - designed SQL import scripts, structured eligibility definitions by family size and income thresholds, automated AMI threshold calculations and recertification workflows.
- Optimized 200+ SQL queries across all client environments (30% performance gain) through execution plan tuning, indexing, partitioning, and audit tables; led database architecture decisions with data architects.
- Eliminated 80% of manual operational workload by designing Python/Bash data pipeline automation; built internal debugging and performance reporting tools reducing debugging time 40%.
- Shipped company's first production AI pipeline on 1.6M financial records - automated data processing workflows replacing manual Excel operations with Python and PyTorch; implemented output validation and quality gates.

PROJECTS

CrashClarity: Large-Scale Road Accident Data Analysis (104K Records)

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages · 2025

- Achieved **85%+ accuracy** on 104K+ UK road accident records by building a classification pipeline (Random Forest + Logistic Regression) with feature engineering across weather, road, temporal, and severity variables.
- Conducted exploratory data analysis, feature selection, cross-validation, and model comparison; deployed interactive analytical dashboard on GitHub Pages for stakeholder reporting.

WHO Global Road Safety Analysis - GMU Research Pipeline

Python · R · ggplot2 · Quarto · Tableau · Statistical Modeling · 2024-2025

- Integrated WHO and UK Road Safety datasets across 190+ countries - data collection, EDA, transformation, and multi-source modeling; produced faculty-facing reports and visualizations in Tableau, ggplot2, and Quarto.
- Implemented Isolation Forest anomaly detection to identify data integrity issues across 3 parallel research pipelines, improving data reliability for downstream faculty analysis.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA: 3.87

Coursework: Statistical Learning, Data Mining, Machine Learning, Database Systems, Data Visualization, Applied Regression

B.Tech Computer Science & Engineering — JNTUA, Anantapur — 2017

SKILLS

Languages & Analysis: Python, R, SQL, PL/SQL, Bash, Excel (pivot tables, VLOOKUP, Power Query)

Data Visualization: Tableau, Power BI, ggplot2, Quarto, Matplotlib, Seaborn, GitHub Pages dashboards

Statistics & ML: Statistical Modeling, Regression Analysis, Hypothesis Testing, Anomaly Detection, Isolation Forest, Random Forest, Logistic Regression, Feature Engineering, Cross-validation, Scikit-learn, NumPy, Pandas

Data Infrastructure: PostgreSQL, MySQL, Oracle, AWS (S3, RDS, Glue), ETL Pipelines, Data Cleaning, Data Transformation, Audit Tables, Database Partitioning

Reporting & Collaboration: Faculty dashboards, Quarto reports, stakeholder documentation, Agile/Scrum, 300+ peer code reviews, cross-functional coordination